



Full Stack Web Developer Program

Go from non-programmer to job-ready web developer — front-end, back-end, databases, essential tools and cloud deployment.

LEARNING PATH · PROGRAM SYLLABUS · REAL-TIME PROJECTS & CASE STUDIES

6 Months

Mon–Fri, 2 Hrs/Day

220 Hrs

Total Program Hours

10+

Real-Time Projects

250+

Hiring Partners

01 Learning Path

1

Cohort Orientation & Preparatory Session

20 Hrs

What full-stack development means, the MEAN vs. MERN stack, web fundamentals, Git/GitHub basics, and basic SQL & programming mathematics.

2

Front-End Development

40 Hrs

Responsive, accessible layouts with HTML5/CSS3/Bootstrap, JavaScript basic-to-advanced, jQuery, DOM manipulation, and building single-page apps with React JS.

3

Back-End Development

40 Hrs

Node.js & Express JS for scalable applications, REST API principles, OOP design patterns, JSON/AJAX for API connectivity, and an overview of Spring Boot & Django.

4

Database Development

40 Hrs

SQL vs. NoSQL concepts, MySQL database creation & manipulation, MongoDB operators/collections/indexes, and PostgreSQL for multi-user web services.

5

Essential Tools

40 Hrs

Git & GitHub for version control, Selenium for web-app testing, Postman for API requests, and project tracking with Trello & JIRA.

6

Deploying Web Applications on Cloud

20 Hrs

AWS smart architecture, deploying a Node.js web server on App Engine, Firebase authentication, and Google Cloud Platform infrastructure.



02 Program Syllabus

TERM 1 · GETTING STARTED

20 HOURS

Cohort Orientation

- What, why & how of full-stack web development
- Frontend, backend, database, testing & version control terms
- Job opportunities and scope for a full-stack developer
- Overview of the MEAN stack and MERN stack

Preparatory Session

- Web fundamentals and Git/GitHub basics
- Basic SQL queries to handle data
- Programming fundamentals & math for programming
- Elite domains: product engineering, data science, DevOps

TOOLS & TECHNOLOGIES

Git

GitHub

VS Code

TERM 2 · FRONT-END DEVELOPMENT

40 HOURS

HTML5, CSS3 & Bootstrap

- Responsive, accessible & appealing layouts
- Bootstrap with the newest HTML5/CSS3 features
- Basics of UI/UX design with Figma
- DOM manipulation techniques

JavaScript & React JS

- JavaScript — basic to advanced, jQuery fundamentals
- Single-page apps with React JS
- Implementing user authentication flows
- Common tags across React, Angular & Node.js apps

TOOLS & TECHNOLOGIES

HTML5

CSS3

Bootstrap

JavaScript

React JS

TERM 2 · BACK-END DEVELOPMENT

40 HOURS

Node.js & Express

- Node JS for scalable, high-performance backends
- Handling HTTP requests using Express JS
- Basic principles of REST APIs
- Overview of Spring Boot and Django

Server-Side Programming Principles

- OOP to design programs using classes & objects
- Design patterns and object modelling
- JSON & AJAX to connect to APIs and endpoints
- Building web applications from scratch (blog app)

TOOLS & TECHNOLOGIES

Node.js

Express.js

REST APIs

**TERM 2 · DATABASE DEVELOPMENT****40 HOURS****SQL & NoSQL**

- SQL vs. NoSQL concepts
- MySQL — creating & manipulating databases
- MongoDB — operators, collections, indexes & cloud
- Built-in interfaces to manage database objects

PostgreSQL

- Producing insights & reports from real-world data
- Handling data warehouses for concurrent web services
- Database object management
- Real use case: On-Demand Home Service Application

TOOLS & TECHNOLOGIES**MySQL****MongoDB****PostgreSQL****TERM 2 · ESSENTIAL TOOLS & CLOUD DEPLOYMENT****60 HOURS****Essential Tools**

- Git & GitHub for version control
- Selenium for quick, systematic web-app testing
- Sending API request collections using Postman
- Project tracking with Trello boards and JIRA

Deploying on Cloud

- Understanding AWS smart architecture
- Writing a Node.js web server ready for App Engine
- Firebase authentication for federated identity
- Google Cloud Platform infrastructure deployment

TOOLS & TECHNOLOGIES**Selenium****Postman****JIRA****AWS****Firebase****GCP**



03 Real-Time Projects & Case Studies

Portfolio Website

Project

List experience & education, showcase projects, SEO-friendly, deployed on GitHub Pages.

Hands-on Build

Website Clone

Project

Pixel-perfect clone of sites like Facebook, Amazon, Flipkart or Zomato across multiple pages.

Hands-on Build

Real-Time Chat Application

Project

Login/signup, invite via email, create chat rooms, and one-on-one messaging.

Hands-on Build

Link Tree

Project

A custom link tree with personal/social links and a QR code, deployed on GitHub Pages.

Hands-on Build

Movie App

Project

Search and view movie details from an online catalogue, deployed on Netlify.

Hands-on Build

Meme Generator

Project

Upload, browse and create trending memes with full login/signup support.

Hands-on Build

Blogging Application

Capstone

Create, edit & delete articles, follow authors, like posts, and add comments with category/tag filters.

Capstone Project

E-Commerce Application

Capstone

User & admin interfaces — browse, filter, review, cart/wishlist, plus inventory and user management.

Capstone Project

Task Management App

Project

Add daily tasks with due dates, mark complete/incomplete, and view weekly/monthly statistics.

Hands-on Build

Library / Stock-Keeping App

Capstone

Browse, rent & review books with a user interface, plus an admin interface to track rentals and send notifications.

Capstone Project